



R3.1 Proton transfer reactions

Reactivity 3 · What are the mechanisms of chemical change? · IB Diploma · Concept guide and original exam-style practice

Understand the concept

According to the Brønsted–Lowry theory, an acid is a proton (H^+) donor and a base is a proton acceptor; in any acid–base reaction, a conjugate acid–base pair differs by a single H^+ . Strong acids/bases dissociate essentially completely in water, while weak acids/bases only partially dissociate, establishing an equilibrium (described by K_a or K_b at HL); the pH scale, $\text{pH} = -\log_{10}[\text{H}^+]$, measures the concentration of hydrogen ions in solution, with pH decreasing as acidity increases, and $\text{pH} + \text{pOH} = 14$ (at 298 K, since $K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14}$). Titration curves for strong/weak acid–base combinations show characteristic shapes, with the equivalence point identified using an appropriate indicator or steepest point of the curve.

Worked example: follow the reasoning

Calculate the pH of a $0.0500 \text{ mol dm}^{-3}$ solution of the strong acid HCl.

Step 1: HCl is a strong (monoprotic) acid, so it dissociates completely: $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$.

Step 2: $[\text{H}^+] = [\text{HCl}] = 0.0500 \text{ mol dm}^{-3}$.

Step 3: $\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10}(0.0500)$.

Step 4: $\text{pH} = -(-1.301) = 1.30$.

Answer: $\text{pH} = 1.30$.

Quick check · multiple choice

In the reaction $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$, which species is the conjugate acid of NH_3 ?

A H_2O

B OH^-

C NH_4^+

D NH_3 itself

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- Define a Brønsted–Lowry acid and a Brønsted–Lowry base. [2] [7]
 - Ethanoic acid, CH_3COOH , is a weak acid. Write an equation showing its dissociation in water, using an equilibrium arrow, and identify the two conjugate acid–base pairs. [3]
 - A solution of ethanoic acid and a solution of hydrochloric acid have the same concentration ($0.100 \text{ mol dm}^{-3}$). State, with a reason, which solution has the higher pH. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) A Brønsted–Lowry acid is a species that donates a proton (H^+) in a reaction; a Brønsted–Lowry base is a species that accepts a proton (H^+). [2]

(b) $\text{CH}_3\text{COOH}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$. Conjugate pairs: $\text{CH}_3\text{COOH}/\text{CH}_3\text{COO}^-$, and $\text{H}_2\text{O}/\text{H}_3\text{O}^+$ (or $\text{H}_3\text{O}^+/\text{H}_2\text{O}$). [3]

(c) The ethanoic acid solution has the higher pH. Ethanoic acid is a weak acid and only partially dissociates in water, giving a lower $[\text{H}^+]$ than hydrochloric acid (a strong acid), which dissociates completely, at the same nominal concentration; lower $[\text{H}^+]$ corresponds to a higher pH. [2]

Quick check answer

C. NH_4^+ is formed when NH_3 (the base) accepts a proton (H^+), so NH_4^+ is the conjugate acid of NH_3 .

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.